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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2024

COMMON TO B.E – CSE / B.TECH – IT & M.TECH - CSE

20MA206 - TRANSFORMS AND ITS APPLICATIONS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** If $L[f(t)] = F(s)$, then $L\left(\frac{f(t)}{t}\right) = \underline{\hspace{2cm}}$.

a) $\int_s^\infty F(s)ds$

b) $\int_0^\infty F(s)ds$

c) $\int_s^0 F(s)ds$

d) $\int_0^\infty F(s)ds$

2. **CO1** Identify the value of $L\left[e^{-3t}t^{\frac{3}{2}}\right]$ is $\underline{\hspace{2cm}}$.

a) $\frac{3\sqrt{\pi}}{4(s+3)^{\frac{5}{2}}}$

b) $\frac{\sqrt{\pi}}{4(s+3)^{\frac{5}{2}}}$

c) $\frac{3\sqrt{\pi}}{4(s+3)^{\frac{3}{2}}}$

d) $\frac{\sqrt{\pi}}{(s+3)^{\frac{5}{2}}}$

3. **CO2** If $F_s(s)$ and $G_s(s)$ are Fourier sine transforms of $f(x)$ and $g(x)$ respectively, then

$\int_0^\infty F_s(s)G_s(s)ds = \underline{\hspace{2cm}}$.

a) $\int_0^\infty f_s(s)g_s(s)ds$

b) $\int_0^\infty f(x)g(x)ds$

c) $\int_0^\infty F_c(s)G_c(s)ds$

d) $\int_0^\infty f(x)g(x)dx$

4. **CO3** The simple poles of $F(z) = \frac{z}{(z-1)(z-2)}$ is $\underline{\hspace{2cm}}$.

a) 1 and 2

b) -1 and 2

c) 1 and -2

d) -1 and -2

5. **CO4** The roots of the differential equation $(D^2 + 4)y = 0$ is _____.
 a) $\pm i$ b) $\pm 2i$ c) $\pm 3i$ d) $\pm 4i$
6. **CO4** The particular integral of the differential equation $(D^2 + 2D - 1)y = e^{2x}$ is _____.
 a) $7e^{2x}$ b) $\frac{1}{7}e^{2x}$ c) $\frac{1}{6}e^{2x}$ d) $\frac{1}{7}e^{3x}$
7. **CO5** $Z[Y_{n+1}] =$ _____.
 a) $z Y(z) - z y(0)$ b) $Y(z) - z y(0)$ c) $zY(z) - y(0)$ d) $z^2 Y(z) - z y(0)$

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO2** The self reciprocal of the Fourier transform $f(x) = e^{-\frac{x^2}{2}}$ is given by _____.
9. **CO3** If $\delta(k)$ is a unit sample function then $Z(\delta(k))$ is _____.
10. **CO5** _____ can be used to solve differential equations and integral equations.

PART - B

(5x 3 = 15 Marks)

11. **CO1** Find $L\{t^3 e^{3t}\}$.
12. **CO2** State and prove change of scale property in Fourier transform.
13. **CO3** State and prove first shifting theorem in Z – transform.
14. **CO4** Solve $(D^3 - 1)y = 0$.
15. **CO4** Solve $(D - 2)^2 y = e^{2x}$.

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Find the Laplace transform of square wave function defined by (7)

$$f(t) = \begin{cases} 1 & 0 < t < a \\ -1 & a < t < 2a \end{cases}, \text{ with period } 2a.$$

- CO1** ii) Find $L^{-1}\left(\frac{1}{(s+a)(s+b)}\right)$ by using Convolution theorem. (8)

17. **CO1** i) Find (i) $L^{-1}\left(\frac{1}{s(s+3)}\right)$ (ii) $L^{-1}\left[\tan^{-1}\left(\frac{s+a}{b}\right)\right]$ (7)

CO2 ii) Show that $e^{-\frac{x^2}{2}}$ is self reciprocal under Fourier Transform. (8)

18. **CO2** i) Find the complex Fourier transform of $f(x)$ if $f(x) = \begin{cases} a-|x| & |x| < a \\ 0 & |x| > a \end{cases}$ and (7)

deduce that $\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$.

CO2 ii) Using Fourier transforms prove that (8)

$$\bullet \int_0^{\infty} \frac{1}{(a^2 + x^2)^2} dx = \frac{\pi}{4a^3}$$

$$\bullet \int_0^{\infty} \frac{x^2 dx}{(a^2 + x^2)(b^2 + x^2)} = \frac{\pi}{2(a+b)}.$$

19. **CO3** i) Find $Z[r^n \cos n\theta]$ and $Z[r^n \sin n\theta]$. (7)

CO3 ii) Find the inverse Z – transform of $\frac{z^2 - 3z}{(z+2)(z-5)}$ by using residues. (8)

20. **CO3** i) Find the inverse Z-Transform of $\frac{z^2}{(z-a)}$ by using Convolution theorem. (7)

CO4 ii) Solve $[(x+1)^2 D^2 + (x+1)D + 1]y = 4 \cos[\log(x+1)]$. (8)

21. **CO4** i) Solve $(x^2 D^2 - xD + 1)y = \left[\frac{\log x}{x}\right]^2$. (7)

CO4 ii) Solve the equation $(D^2 - 4D + 3)y = \sin 3x + x^2$. (8)

22. **CO5** i) Solve $y'' + 4y' - 5y = 5$ given that $y = 0$ and $y' = 2$ when $t = 0$ using Laplace transform. (7)

CO5 ii) Solve $y_{n+2} - 4y_{n+1} + y_n = 0$, $y(0) = 1$, $y(1) = 0$ using Z transforms. (8)
